

1. Introduction to Databases

Introduction to databases

Exercise:

List 3 real world systems you would store in a database.

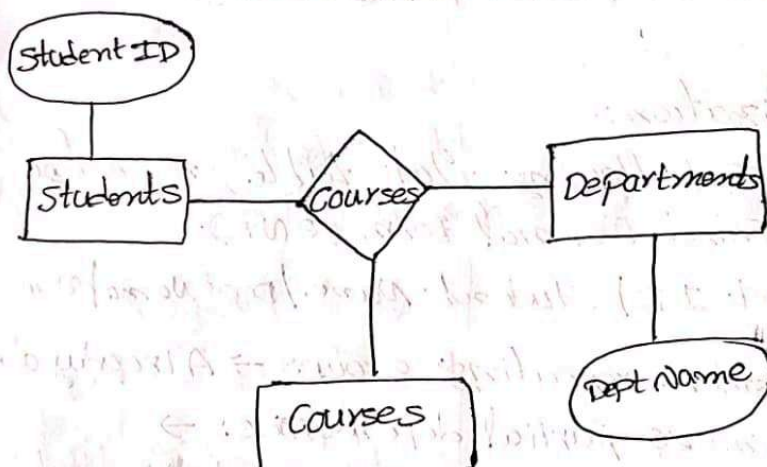
School Database → students, Teachers, Admin, Fee, Exams, Progress.

Hospital Database → Patients, doctors, appointments, billing.

E-commerce Database → Customers, products, orders, payments.

2. E-R Modeling & Diagrams

ER modeling & Diagrams:-



3. ACID Properties

Acid properties:-

Banking transaction: Transfer 1000 from Account A to Account B.

Transfer $A \rightarrow B$

- Atomicity: Either both debit and Credit should happen, or neither.
- Consistency: Total Money before and after transaction same.
- Isolation: If A transfer to two people at a same time, It will allow to only one people.
- Durability: Once transaction done, the transfer is stored permanently.

4. Normalization

4) Normalization:-

Given the following flat table, normalize it to Third Normal Form (3NF):

(Student ID) Student Name / Dept Name / Advisor

1NF : Remove repeating groups → Already atomic

2NF : Remove partial dependencies →
Introduce Departments and Advisor tables

3NF : Remove Transitive dependencies → Advisor depends on Dept,

Tables:

Students (Student ID, Student Name, Dept ID)

Departments (Dept ID, Dept Name, Advisor ID)

Advisors (Advisor ID, Advisor Name).